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XIAMEN
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CST 205

DIGITAL LOGIC

Synchronous Sequential Logic Circuit

Chapter 5.3

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5.3.2 State reduction

What is state reduction? The so-called state reduction refers to the use of some reduction technique to eliminate redundant state from the basic state table, and obtain a state table that can correctly describe a given logic function and minimize the number of states involved. This state table is called a modified state table or reduced state table.

Objective: Simplify the circuit structure. The number of states directly determines how many flip-flops are needed in the circuit. Let the number of states be n and the number of flip-flops required be m , then the following relationship should be satisfied:

$$2^m \geq n > 2^{m-1}$$

Methods: Common methods include observation method, output classification method, and implicant chart method. The most commonly used methods are discussed below – the observation method and implication chart method.



I. 、 Reduction of completely specified state table

1. Concepts

(1) State equivalence

Please pay attention to three points: definition, determine methods and properties.

①Definition

Assume the states S_i and S_j are the two states of the completely specified table; for all the possible input combinations, all the response sequence from states S_i and S_j are fully identical. Then, we can say that **states** S_i and S_j are equivalent, written as (S_i, S_j) , it is also called **states** S_i and S_j as one equivalent pair.



② Determine method

For all the input combinations, S_i and S_j are the two present states of the fully complete basic state table, the conditions that S_i and S_j are equivalent to each other, must satisfy the following two conditions:

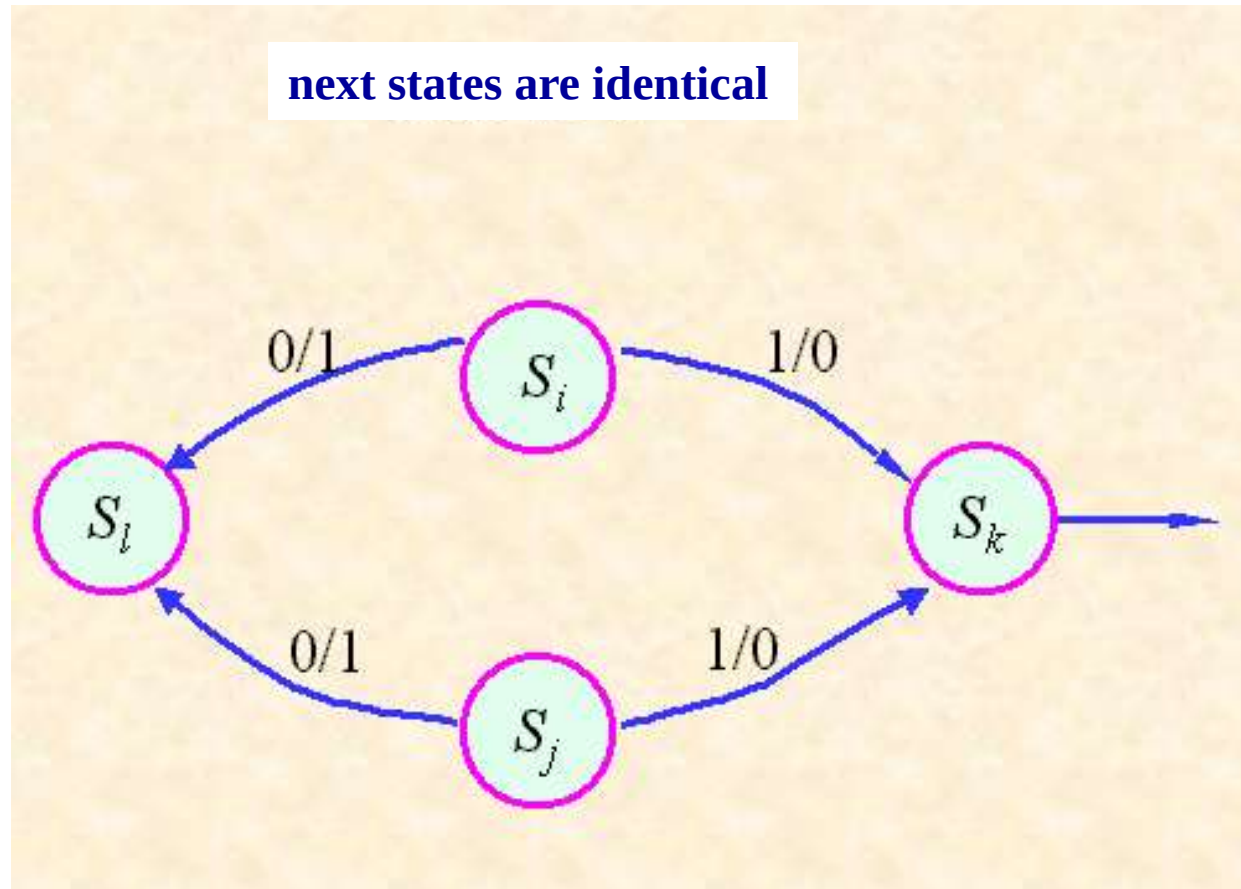
First , the output must be identical;

Second, the next state must belong to one of the following sub-conditions:

- a. the next states are identical ;
- b. the next states are symmetric;
- c. the next states are cyclic or points to an equivalent pare.

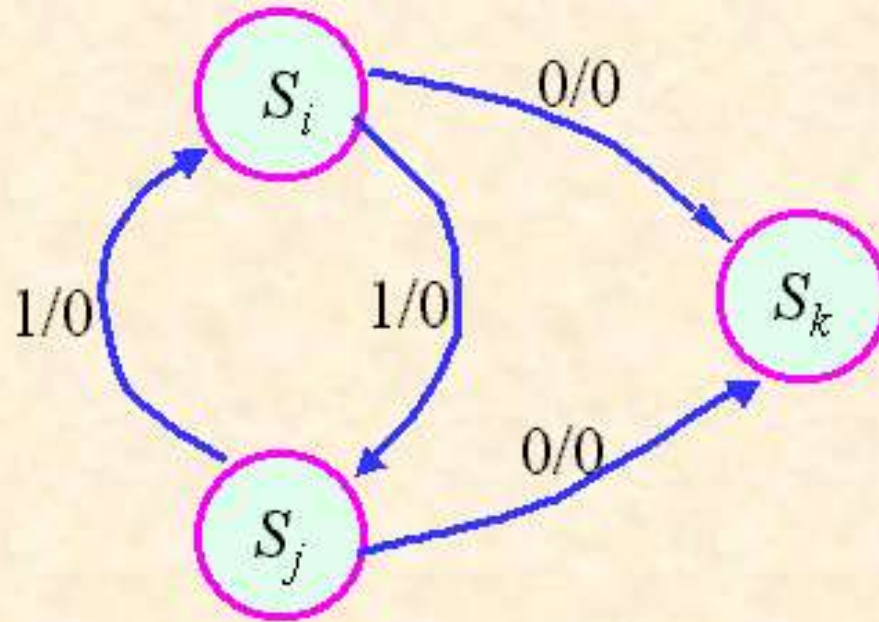


The next states are identical



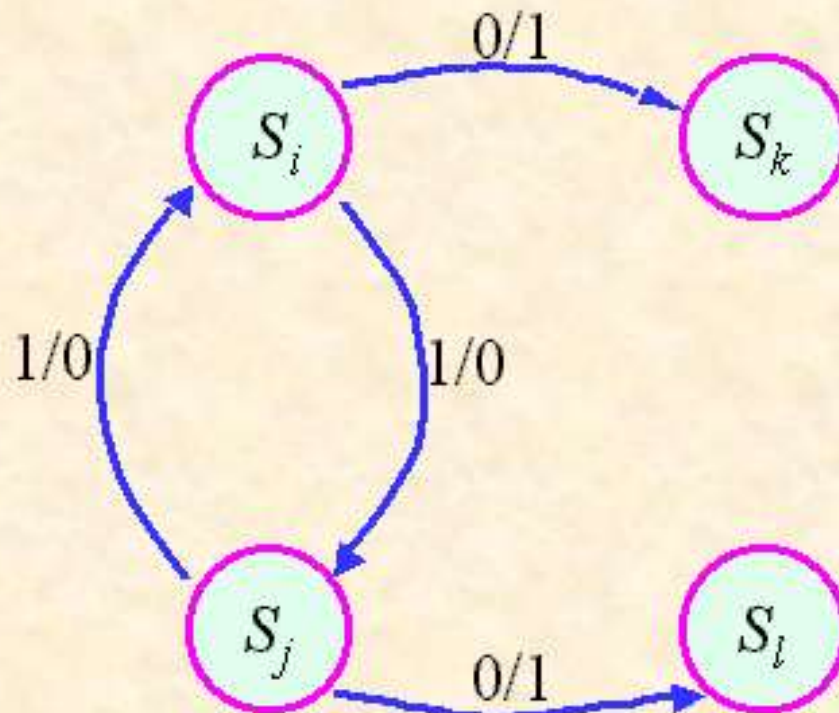
The next states are symmetric

next states are symmetric
or identical



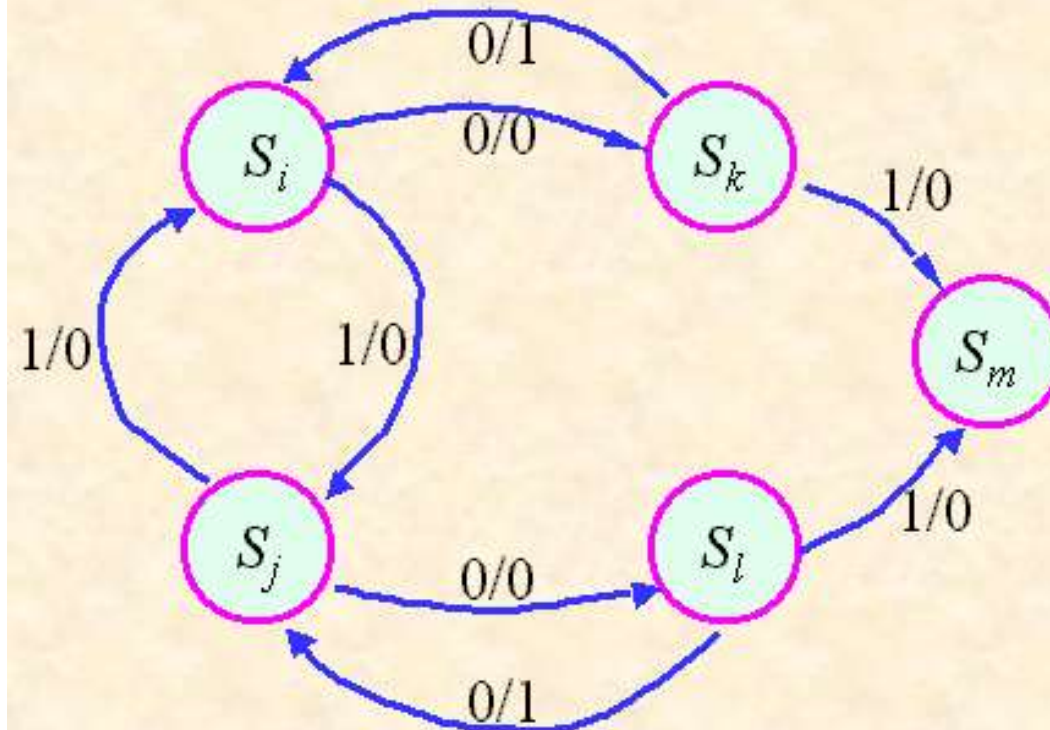
The next states are symmetric

next states are symmetric or identical
(S_k and S_l are equivalent)



The next states are cyclic or points to an equivalent pair

next states are symmetric, cyclic or points to an equivalent pair



③ Properties:

Equivalent state has the transitivity, i.e., if S_1 and S_2 are equivalent, as well as, S_2 and S_3 are equivalent, then, S_1 and S_3 must be equivalent, written as:

$$(S_1, S_2), (S_2, S_3) \longrightarrow (S_1, S_3)$$

(2)Equivalent class

Equivalent class: A collection of states that are equivalent to each other. Any two states in the same equivalent class are equivalent. E.g, in term of (S_1, S_2) and (S_2, S_3) , we can have (S_1, S_3) , so that:

S_1 、 S_2 、 S_3 belongs to one class of state, written as $\{ S_1, S_2, S_3 \}$, i.e.,

$$(S_1, S_2), (S_2, S_3) \longrightarrow \{ S_1, S_2, S_3 \}$$



(3) maximum equivalent class

The so-called maximum equivalent class refers to the equivalent class that is not included in any other equivalent class.

In other words, if an equivalent class is not a subset of any other equivalent class, then the equivalent class is called the largest equivalent class.

The reduction process of the basic state table is to find all the maximum equivalent classes in the table, and then merge the states in each of the maximum equivalent classes into one new state, thus obtaining a minimized state table.

The number of states after reduction is equal to the number of maximum equivalent classes !



2. State reduction

(1) Observation method

Example: Reduce the basic state table

Present state	Next state/Output	
	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	D/1
D	A/0	D/1



Example: Reduce the basic state table

Present state	Next state/Output	
	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	D/1
D	A/0	D/1

- Identical output:
 - ▣ A and B, C and D
- For all the input combinations, C and D have the same next states. Thus, C and D are equivalent.
- For $X=1$, the next states of A and B cannot satisfy the three subconditions. Thus, A and B are NOT equivalent.
- The maximum equivalent classes are {A}, {B}, {C, D},

2. 2. State reduction

(1) General steps for implication chart method

Draw unfilled implication chart

①

Find the state equivalent pares

②

Obtain maximum equivalent classes

③

Obtain modified state table

⑤

State merge

④

① Draw unfilled implication chart

The implicant chart is a rectangular triangle step-like grid, each square in the table represents a state pair.



② Find the state equivalent pares

Sequence comparison: according to the order of the implicant chart from the upper to the lower, from the left to the right, all "state pairs" are checked and compared one by one in turn from the basic state table, and the results are labeled in the corresponding squares in the implicant chart.

The comparison results are labeled as follows:

Equivalent ----- Fill “ $\vee$ ” in the corresponding square;

Inequivalent----- Fill “ $\times$ ” in the corresponding square;

Rely on other state pares---- Fill the relied state pares in the corresponding square.

Correlation comparison: Refer to the state check for those pairs, which have not been determined to be equivalent, at the time of sequence comparison. Until their relied state pares can be determined, we can determine their states are equivalent or not equivalent.

③ Obtain maximum equivalent classes

After finding all the equivalent pairs in the basic state table, we can use the transitivity of the equivalent states to find the maximum equivalent class.

④ State merge, make a modified state table

Combine all the states in each of the maximum equivalent classes into one state to get a modified state table that is equivalent to the basic state table.



(2)Example:

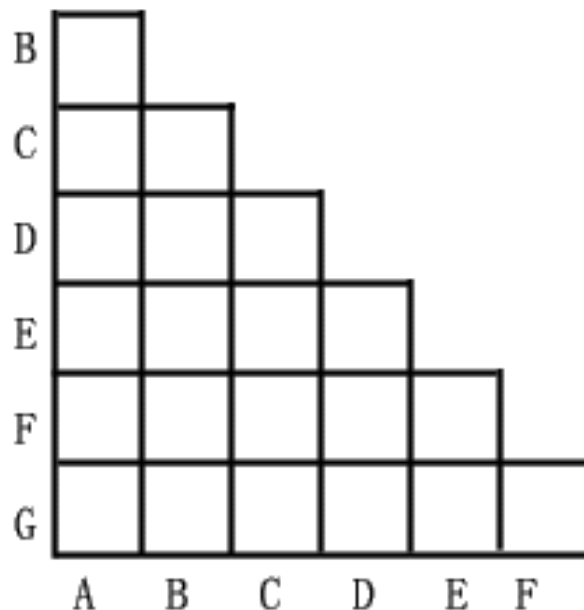
Q: Reduce the following basic state table

basic state table

Present State	Next state/output	
	x=0	x=1
A	C/0	B/1
B	F/0	A/1
C	F/0	G/0
D	D/1	E/0
E	C/0	E/1
F	C/0	G/0
G	C/1	D/0

A: ① Draw unfilled implication chart

The given basic state table has 7 states, according to the rules for filling implicant chart, the unfilled implicant chart frame can be drawn as follows.



② Find the state equivalent pares

According to the judgment standard of the equivalent state, each state pair is checked in turn, and the result of the sequence comparison can be obtained as shown in Fig (a).

The results of the correlation comparison are shown in Fig (b)

basic state table

B	CF					
C	X	X				
D	X	X	X			
E	BE	AE CF	X	X		
F	X	X	✓	X	X	
G	X	X	X	CD DE	X	X
	A	B	C	D	E	F

Fig (a)

B	CF					
C	X	X				
D	X	X	X			
E	BE	AE CF	X	X		
F	X	X	✓	X	X	
G	X	X	X	CD DE	X	X
	A	B	C	D	E	F

Fig (b)

Present State	Next state/output	
	x=0	x=1
A	C/0	B/1
B	F/0	A/1
C	F/0	G/0
D	D/1	E/0
E	C/0	E/1
F	C/0	G/0
G	C/1	D/0



B	CF					
C	X	X				
D	X	X	X			
E	BE	AE CF	X	X		
F	X	X	✓	X	X	
G	X	X	X	CF DE	X	X
	A	B	C	D	E	F

It can be seen from the result that the seven states in the basic state table have four equivalent pairs:

(A, B), (A, E), (B, E), (C, F)。



③ Obtain the maximum equivalent classes

In terms of the obtained equivalent pairs and the definition of the maximum equivalent class, the seven states can merge to four maximum equivalent classes, $\{A,B,E\}, \{C,F\}, \{D\}, \{G\}$.

④ Merge the states and obtain the modified state table

Let $\{A,B,E\}$ ----a 、 $\{C,F\}$ ---b 、 $\{D\}$ ---c 、 $\{G\}$ ---d , substitute them into the basic state table, then we can obtain the modified state table:

Basic state table

Present State	Next state/output	
	x=0	x=1
A	C/0	B/1
B	F/0	A/1
C	F/0	G/0
D	D/1	E/0
E	C/0	E/1
F	C/0	G/0
G	C/1	D/0



Modified state table

Present State	Next state/output	
	x=0	x=1
a	b/0	a/1
b	b/0	d/0
c	c/1	a/0
d	b/1	c/0



Practise-State Reduction with the implicant chart



Q_n	x	0	1
A		D,0	B,0
B		D,0	C,0
C		D,0	C,1
D		D,0	B,0

Practise-Answer



Q_n	x	0	1
A		D,0	B,0
B		D,0	C,0
C		D,0	C,1
D		D,0	B,0

B	BC		
C	×	×	
D	✓	BC	×
	A	B	C

II. Reduction of the incomplete specified state

The reduction of the incomplete specified state introduces the concept of a compatible state. See the textbook for details!



5.3.3 State assignment

State assignment: refers to the state indicated by letters or numbers in the reduce state table, specifying a binary code to form a binary state table. State assignment are also called state coding.

The task of the state assignment:

① Determine the length of the state code (ie the number of bits in the binary code, or the required number of flip-flops);

② Look for an optimal or secondary-optimal state allocation scheme, so as to make the designed sequential circuit the easiest.



1. Determine the number of bits in the binary code

Let the state of the modified state table be N , and the length of the state code be m , then the relationship between the state number N and the state code length m is

$$2^{m-1} < N \leq 2^m$$

For example, if the state number of a state table is $N = 7$, the number of bits in the binary code should be $m = 3$ when the state is assigned. Or the number of state variables is 3.

II. Determine the state assignment scheme

There are many kinds of correspondences between states and codes. In general, the number of state allocation schemes that may occur when allocating N states with 2^m combinations of m -bit binary codes, K_s is

$$K_s = A_{2^m}^N = \frac{2^m!}{(2^m - N)!}$$

eg, when $N = 4$,
 $m = 2$, $K_s = 24$.



Example: when $n=2$, $N=4$, we have

States	solutions											
	1	2	3	4	5	6	7	8	9	10	11	12
A	00	10	01	11	00	01	10	11	00	10	01	11
B	01	11	00	10	10	11	00	01	11	01	10	00
C	11	01	10	00	11	10	01	00	01	11	00	10
D	10	00	11	01	01	00	11	10	10	00	11	01

States	solutions											
	13	14	15	16	17	18	19	20	21	22	23	24
A	00	01	10	11	00	10	01	11	00	01	10	11
B	11	10	01	00	10	00	11	01	01	00	11	10
C	10	11	00	01	01	11	00	10	10	11	00	01
D	01	00	11	10	11	01	10	00	11	10	01	00

A common method is called the adjacent assignment method.

The basic idea of the adjacent assignment method is to make the excitation function and the output function in the adjacent position of the "1" square on the Karnaugh map as much as possible, so as to facilitate the excitation function and the output function. simple.



The principle principle of the adjacent assignment method is as follows:

① If the next state is the same, then the present state should be adjacent. (ie, under the same input conditions, the present state with the same next state should be allocated adjacent binary code as much as possible;)

② For the same present states, their next states should be adjacent. (ie, under the adjacent input conditions, the next state of the same present state should be assigned adjacent code as much as possible ;)

③ If the output is the same, then the present states should be adjacent. (ie, the present state with the same output for each input value should be assigned to the adjacent binary code as much as possible .)

Some state tables often have situations where three principles cannot be met at the same time. At this time, it can be considered in order of priority from ① to ③.

In addition, the initial state is generally assigned the "0" state in consideration of the actual operating state of the circuit. 

III. Example

Q: Assign binary code for the following state table (set A to the initial state).

Present state	Next state/Output	
	x=0	x=1
A	C/1	B/0
B	A/0	A/1
C	A/1	D/1
D	D/1	C/0

A: In the state table shown, the number of states is $N = 4$, so the length of the state code should be $m=2$. That is, the function of implementing the state table requires two flip-flops.

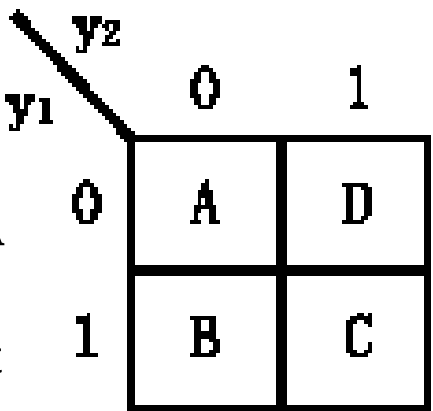


According to the code assignment principle of the adjacent method, the adjacent relationship of the four states is as follows:

Based on Rule ①, States B and C should be assigned the adjacent binary codes;
Based on Rule ②, States B and C, A and D, C and D should be assigned the adjacent binary codes ;
Based on Rule ③, State A and D should be assigned the adjacent binary codes .
Collectively considering ① ~③ , State assignments are required to satisfy B and C, A and D, C and D adjacent.
In order to intuitively assign the codes, the Karnaugh map is usually used as a tool for state assignment.

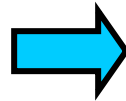
Assume states are presented by y_2y_1 , And assign A to "0", an assignment scheme that satisfies the above-mentioned neighbor relationship as shown in the right figure. That is, the state codes of states A, B, C, and D are sequentially 00, 01, 11, and 10 of y_2y_1 .

Present state	Next state/Output	
	x=0	x=1
A	C/1	B/0
B	A/0	A/1
C	A/1	D/1
D	D/1	C/0



The states A, B, C, and D in the state table are replaced by codes 00, 01, 11, and 10, respectively, to obtain a binary state table of the state table as shown in the table below.

Present state	Next state/Output	
	x=0	x=1
A	C/1	B/0
B	A/0	A/1
C	A/1	D/1
D	D/1	C/0



Present state y_2y_1	Next state/ $y_2^{(n+1)}y_1^{(n+1)}$ /Output	
	x=0	x=1
00	11/0	01/0
01	00/0	00/1
11	00/1	10/1
10	10/0	11/0

Note: There are usually many options for meeting the assignment principle, and the designer can choose one of them.



Practice

Example: Assign binary states for the following state table.

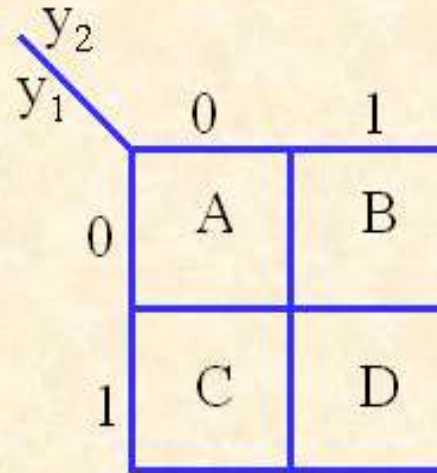
Present State	next State/output	
	$x=0$	$x=1$
A	C/0	D/0
B	C/0	A/0
C	B/0	D/0
D	A/1	B/1

Answer:

Based on Rule ①, States A and B, A and C, should be assigned the adjacent binary codes;

Based on Rule ②, States C and D, C and A, B and D, A and B should be assigned the adjacent binary codes ;

Based on Rule ③, State A, B and C should be assigned the adjacent binary codes . i.e., State A and B, A and C, B and C should be adjacent.



5.3.4 Determine the excitation and output functions and draw a logic diagram

Task: According to the binary state table and the excitation table of the selected flip-flops, find the excitation function expression of the flip-flops and the output function expression of the circuit, and simplify it; so as to construct a logic circuit that implements a given logic function with appropriate logic gates and selected flip-flops.



Flip-flop excitation table

- The excitation table of a flip-flop device illustrates the flip-flop reacts to the input (excitation) from a present state to a next state. In this table, the present state and the next state are used as independent variables, and the input is used as the dependent variables. The excitation table for the flip-flop can be derived from the flip-flop's function table.

Flip-flop excitation table

R-S Flip-flop excitation table

Q	$Q^{(n+1)}$	R	S
0	0	d	0
0	1	0	1
1	0	1	0
1	1	0	d

D Flip-flop excitation table

Q	$Q^{(n+1)}$	D
0	0	0
0	1	1
1	0	0
1	1	1

Flip-flop excitation table

J-K Flip-flop excitation table

Q	$Q^{(n+1)}$	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

T Flip-flop excitation table

Q	$Q^{(n+1)}$	T
0	0	0
0	1	1
1	0	1
1	1	0

According to the binary state table and the flip-flop excitation table, the simplest expression for the excitation function and the output function is generally divided into two steps:

- List the excitation and output function truth tables;
- Write the simplest expression after simplifying the Karnaugh maps.

Example: The following binary state table functions are implemented with J-K flip-flops and appropriate logic gates.

Present state y_2y_1	Next state $y_2^{(n+1)}y_1^{(n+1)}/\text{Output}$	
	$x=0$	$x=1$
00	11/0	01/0
01	00/0	00/1
10	01/0	11/0
11	00/1	10/1



A: The truth table for the excitation and output functions can be listed according to the given binary state table and the excitation table for the J-K flip-flop as shown in the table to the right.

Present state $y_2 y_1$	Next state $y_2^{(n+1)} y_1^{(n+1)}$ / Output	
	$x=0$	$x=1$
00	11/0	01/0
01	00/0	00/1
10	01/0	11/0
11	00/1	10/1

$Q Q^{(n+1)}$	J K
0 0	0 d
0 1	1 d
1 0	d 1
1 1	d 0

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$				Output Z
0	0	0	1	d	1	d	0
0	0	1	0	d	d	1	0
0	1	0	d	1	1	d	0
0	1	1	d	1	d	1	1
1	0	0	0	d	1	d	0
1	0	1	0	d	d	1	1
1	1	0	d	0	1	d	0
1	1	1	d	0	d	1	1



YZ

Excitation and Output function truth table

$QQ^{(n+1)}$

J K

0 0

0 d

0 1

1 d

1 0

d 1

1 1

d 0

Pre $y_2 y_1$	Next $y_2^{(n+1)} y_1^{(n+1)}$ / output	
	x=0	x=1
00	11/0	01/0
01	00/0	00/1
10	01/0	11/0
11	00/1	10/1

x	y_2	y_1	Next state $y_2^{(n+1)} y_1^{(n+1)}$		Excitation : $J_2 K_2 J_1 K_1$				Output Z
			$y_2^{(n+1)}$	$y_1^{(n+1)}$	J_2	K_2	J_1	K_1	
0	0	0	1	1	1	d	1	d	0
0	0	1	0	0	0	d	d	1	0
0	1	0	0	1	d	1	1	d	0
0	1	1	0	0	d	1	d	1	1
1	0	0	0	1	0	d	1	d	0
1	0	1	0	0	0	d	d	1	1
1	1	0	1	1	d	0	1	d	0
1	1	1	1	0	d	0	d	1	1

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

$x y_2$	00	01	11	10
y_1 0	1	d	d	0
1	0	d	d	0

$$J_2 = \bar{x} \cdot \bar{y}_1$$

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

$x y_2$	00	01	11	10
y_1 0	d	1	0	d
1	d	1	0	d

$$K_2 = \bar{x}$$

Excitation and Output function truth table

x	y_2	y_1	Excitation				Output
			J_2	K_2	J_1	K_1	Z
0	0	0	1	d	1	d	0
0	0	1	0	d	d	1	0
0	1	0	d	1	1	d	0
0	1	1	d	1	d	1	1
1	0	0	0	d	1	d	0
1	0	1	0	d	d	1	1
1	1	0	d	0	1	d	0
1	1	1	d	0	d	1	1

xy_2		00	01	11	10
y_1	0	1	1	1	1
	1	d	d	d	d

$$J_1 = 1$$

Excitation and Output function truth table

x	y_2	y_1	Excitation $J_2 K_2 J_1 K_1$	Output Z
0	0	0	1 d 1 d	0
0	0	1	0 d d 1	0
0	1	0	d 1 1 d	0
0	1	1	d 1 d 1	1
1	0	0	0 d 1 d	0
1	0	1	0 d d 1	1
1	1	0	d 0 1 d	0
1	1	1	d 0 d 1	1

$x y_2$	00	01	11	10
y_1				
0	d	d	d	d
1	1	1	1	1

$$K_1=1$$

Excitation and Output function truth table

x	y_2	y_1	Excitation				Output Z
			J_2	K_2	J_1	K_1	
0	0	0	1	d	1	d	0
0	0	1	0	d	d	1	0
0	1	0	d	1	1	d	0
0	1	1	d	1	d	1	1
1	0	0	0	d	1	d	0
1	0	1	0	d	d	1	1
1	1	0	d	0	1	d	0
1	1	1	d	0	d	1	1

xy_1	xyz			
	00	01	11	10
0	0	0	0	0
1	0	1	1	1

Z

$$Z = y_2 y_1 + x y_1 = (y_2 + x) y_1$$

Based on the Karnaugh, the truth table can be used to make the excitation and output functions is shown below.

$\overline{x}y_2$	00	01	11	10
y_1 0	1	d	d	0
1	0	d	d	0

$$J_2 = \overline{x} \cdot \overline{y_1}$$

$\overline{x}y_2$	00	01	11	10
y_1 0	d	1	0	d
1	d	1	0	d

$$K_2 = \overline{x}$$

$\overline{x}y_2$	00	01	11	10
y_1 0	1	1	1	1
1	d	d	d	d

$$J_1 = 1$$

$\overline{x}y_2$	00	01	11	10
y_1 0	d	d	d	d
1	1	1	1	1

$$K_1 = 1$$

$\overline{x}y_2$	00	01	11	10
y_1 0	0	0	0	0
1	0	1	1	1

$$Z$$

$$Z = y_2 y_1 + x y_1 = (y_2 + x) y_1$$

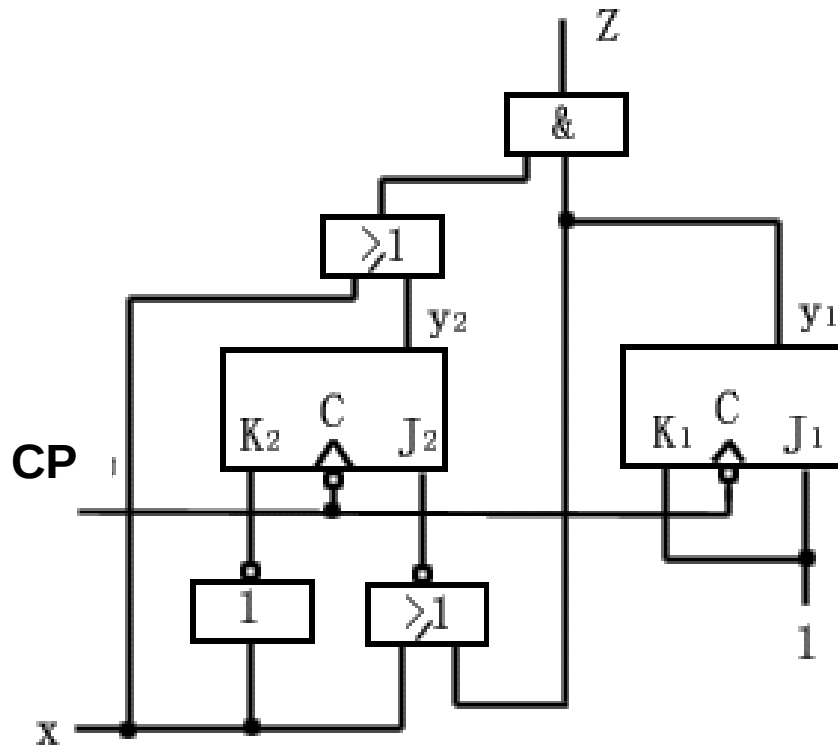


The simplified expressions of the excitation and output functions obtained by simplification are as follows:

$$J_2 = \overline{x}y_1 ; \quad J_1 = K_1 = 1$$

$$K_2 = \overline{x} ; \quad Z = y_2y_1 + xy_1 = (y_2 + x) y_1$$

The corresponding logic circuit diagram is shown below.



Example: Using the JK flip-flops to implement the binary state table shown in the table below, try to write the excitation and output functions.

$y_1 y_2 \backslash x$	0	1
00	10,0	11,0
01	10,0	00,0
10	01,0	11,0
11	00,1	01,1

$Q Q^{(n+1)}$	J K
0 0	0 d
0 1	1 d
1 0	d 1
1 1	d 0

$y_1 y_2 \backslash x$	0	1
$J_2 K_2 \ J_1 K_1$	$J_2 K_2 \ J_1 K_1$	$J_2 K_2 \ J_1 K_1$
00	1d,0d	1d,1d
01	1d,d1	0d,d1
10	d1,1d	d0,1d
11	d1,d1	d1,d0

$y_1 y_2 \backslash x$	0	1
00	0	0
01	0	0
10	0	0
11	1	1

J_2

$y_1 y_2$	00	01	11	10
x				
0	1	1	d	d
1	1	0	d	d

$$J_2 = \bar{x} + \bar{y}_2$$

K_2

$y_1 y_2$	00	01	11	10
x				
0	d	d	1	1
1	d	d	1	0

$$K_2 = \bar{x} + y_2$$

J_1

$y_1 y_2$	00	01	11	10
x				
0	0	d	d	1
1	1	d	d	1

$$J_1 = x + y_1$$

K_1

$y_1 y_2$	00	01	11	10
x				
0	d	1	1	d
1	d	1	0	d

$$K_1 = \bar{x} + \bar{y}_1$$

Z

$y_1 y_2$	00	01	11	10
x				
0	0	0	1	0
1	0	0	1	0

$$Z = y_1 y_2$$

$y_1 y_2 \backslash x$	0	1
	$J_2 K_2 \ J_1 K_1$	$J_2 K_2 \ J_1 K_1$
00	1d,0d	1d,1d
01	1d,d1	0d,d1
10	d1,1d	d0,1d
11	d1,d1	d1,d0

Example: Use the D flip-flops to implement the binary state table, please list the excitation and output functions

Present state		Next state $y_2^{(n+1)}y_1^{(n+1)}$ / Output Z	
y_2	y_1	$x=0$	$x=1$
0	0	10/0	01/0
0	1	00/0	00/1
1	1	01/1	10/1
1	0	00/1	11/1

A: Determine the excitation function

Q	$Q^{(n+1)}$	D
0	0	0
0	1	1
1	0	0
1	1	1

Input	Present		Next		Excitation	
x	y_2	y_1	$y_2^{(n+1)}$	$y_1^{(n+1)}$	D_2	D_1
0	0	0	1	0	1	0
0	0	1	0	0	0	0
0	1	1	0	1	0	1
0	1	0	0	0	0	0
1	0	0	0	1	0	1
1	0	1	0	0	0	0
1	1	1	1	0	1	0
1	1	0	1	1	1	1

		xy_2			
y_1		00	01	11	10
	0	1	0	1	0
	1	0	0	1	0

D_2

		xy_2			
y_1		00	01	11	10
	0	0	0	1	1
	1	0	1	0	0

D_1

Determine the
output function

Present
state

$y_2^{(n+1)}y_1^{(n+1)}/$

Output Z

$y_2 \ y_1$	$x=0$	$x=1$
0 0	10/0	01/0
0 1	00/0	00/1
1 1	01/1	10/1
1 0	00/1	11/1

$y_2 \backslash y_1$	00	01	11	10
0	0	1	1	0
1	0	1	1	1

Z

$$Z = y_2 + xy_1$$

5.4 Synchronous sequential logic circuit design example

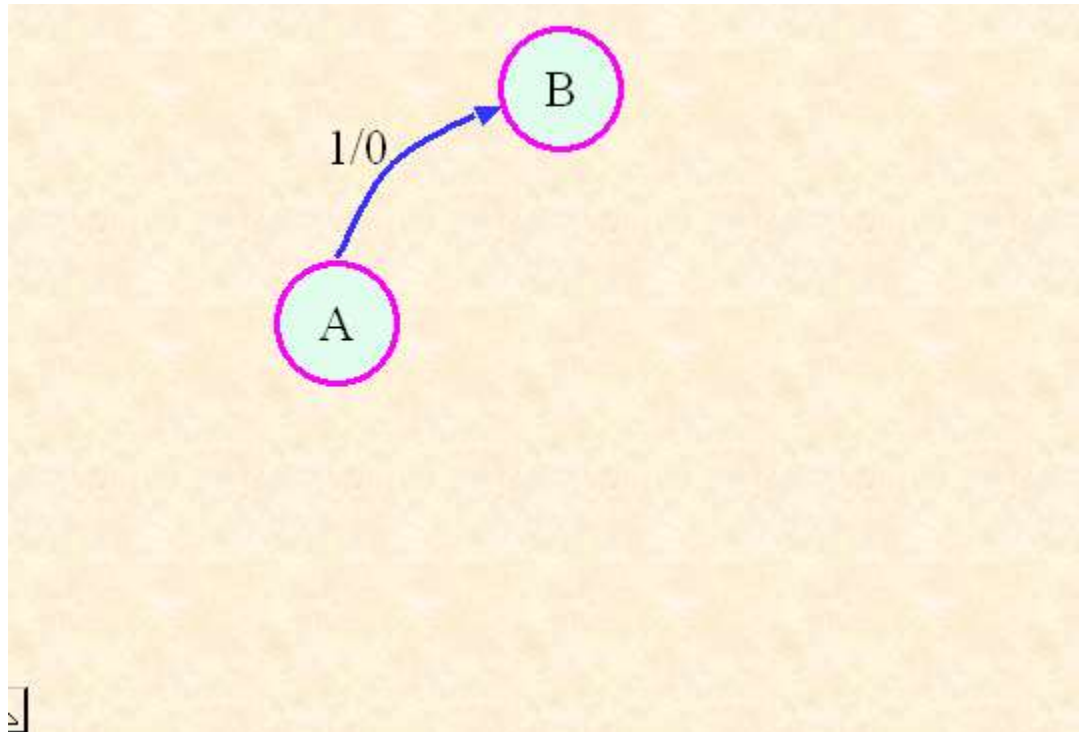
Example:

Design a “111” sequence detector, to check a sequence of binary numbers. When we have three continuous 1s, the output is 1; otherwise, the output is 0. the typical input/output are list as follows:

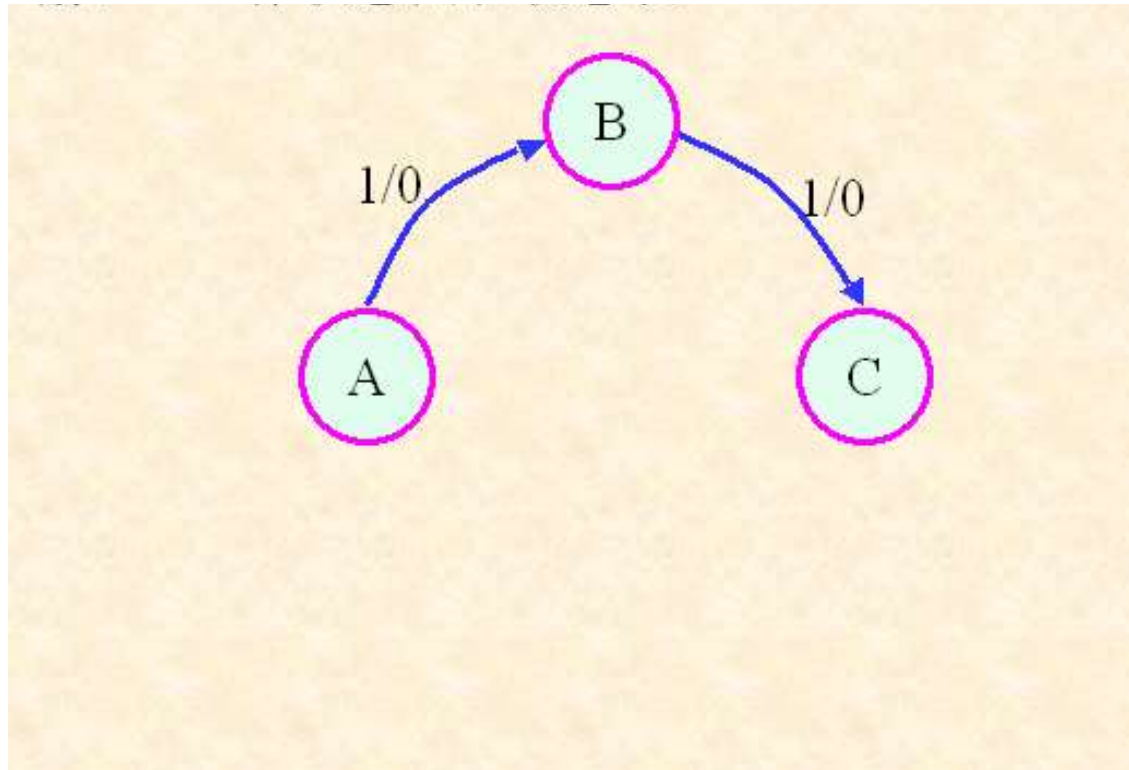
Input	X:	0	1	1	1	0	1	1	1	1	0
Output	Z:	0	0	0	1	0	0	0	1	1	0



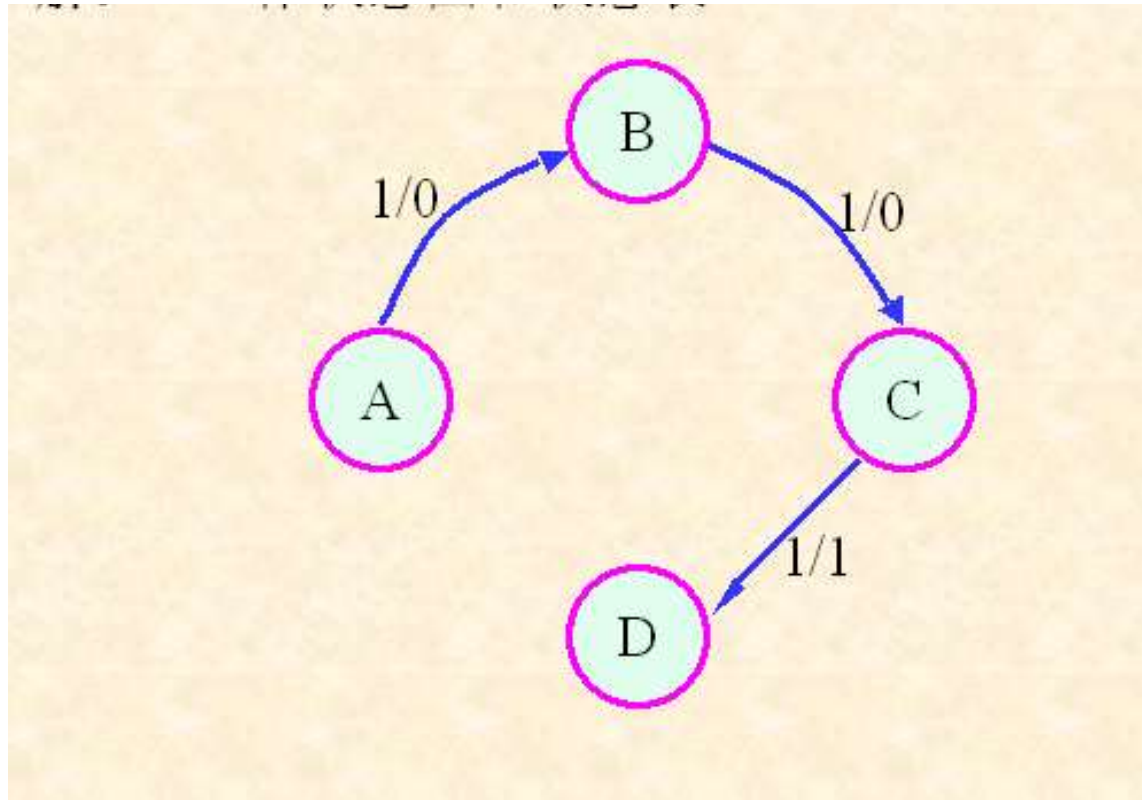
A: basic state diagram and state table



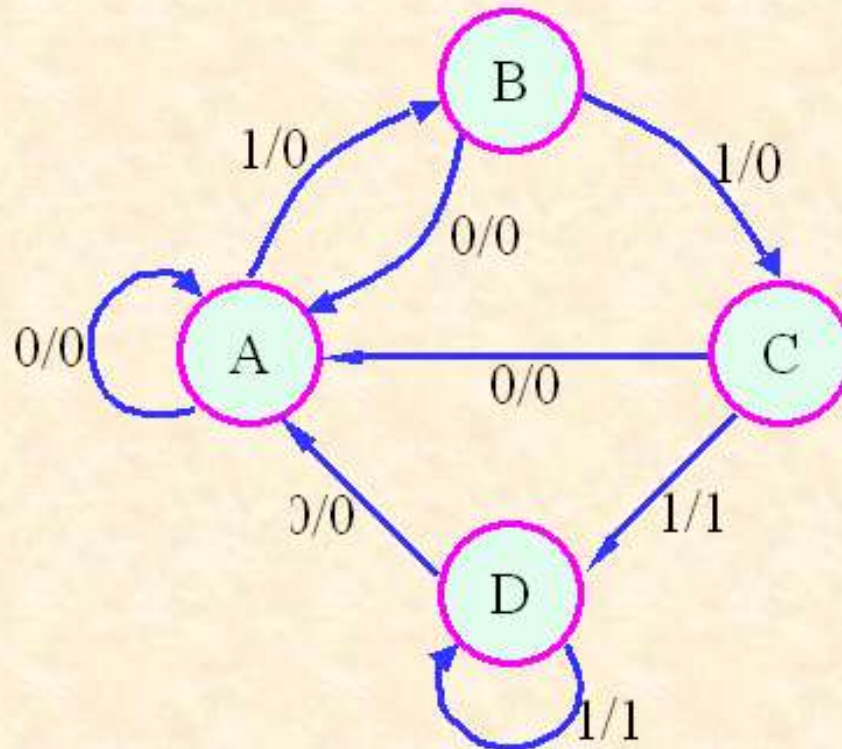
A: basic state diagram and state table



A: **basic state diagram and state table**



A: basic state diagram and state table



A: basic state diagram and state table

Present State	Next state/Output Z	
	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	D/1
D	A/0	D/1

- State reduction

The maximum equivalent classes are: {A}, {B}, {C,D}

Let C to represent {C,D}, we can have the following table

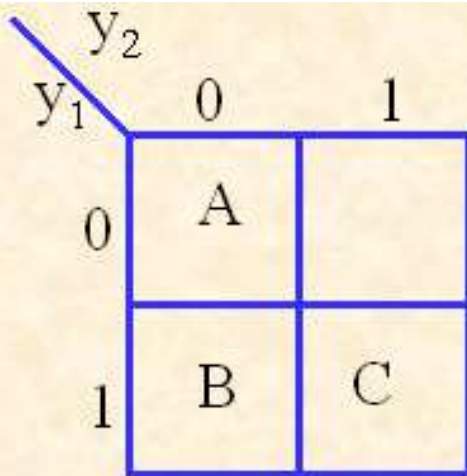
Present State	Next state/Output Z		Present State	Next state/Output Z	
	$x=0$	$x=1$		$x=0$	$x=1$
A	A/0	B/0	A	A/0	B/0
B	A/0	C/0	B	A/0	C/0
C	A/0	D/1	C	A/0	C/1
D	A/0	D/1			



State Assignment:

Present State	Next state/Output Z	
	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	C/1

- A and B, B and C, A and C should be adjacent;
- A and B, A and C should be adjacent;
- A and B should be adjacent;
- A should be assigned as 0;



Present State

Next state/Output Z

$y_2 \ y_1$		$x=0$	$x=1$
0	0	00/0	01/0
0	1	00/0	11/0
1	1	00/0	11/1

Present State

Next state/Output Z

	$x=0$	$x=1$
A	A/0	B/0
B	A/0	C/0
C	A/0	C/1

Determine the excitation function and output function expressions;

Use the JK flip-flops

Q	Q ⁽ⁿ⁺¹⁾	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

Input	Present		Next state		Excitation				Output Z
x	y ₂	y ₁	y ₂ ⁽ⁿ⁺¹⁾	y ₁ ⁽ⁿ⁺¹⁾	J ₂	K ₂	J ₁	K ₁	Z
0	0	0	0	0	0	d	0	d	0
0	0	1	0	0	0	d	d	1	0
0	1	1	0	0	d	1	d	1	0
0	1	0	d	d	d	d	d	d	d
1	0	0	0	1	0	d	1	d	0
1	0	1	1	1	1	d	d	0	0
1	1	1	1	1	d	0	d	0	1
1	1	0	d	d	d	d	d	d	d

Input Present Excitation Output Z

x	y ₂	y ₁	J ₂	K ₂	J ₁	K ₁	Z
0	0	0	0	d	0	d	0
0	0	1	0	d	d	1	0
0	1	1	d	1	d	1	0
0	1	0	d	d	d	d	d
1	0	0	0	d	1	d	0
1	0	1	1	d	d	0	0
1	1	1	d	0	d	0	1
1	1	0	d	d	d	d	d

	xy ₂ 00	01	11	10
y ₁				
0	0	d	d	0
1	0	0	1	0

	xy ₂ 00	01	11	10
y ₁				
0	0	d	d	0
J ₂				
1	0	d	d	1

	xy ₂ 00	01	11	10
y ₁				
0	d	d	d	d
K ₂				
1	d	1	0	d

	xy ₂ 00	01	11	10
y ₁				
0	0	d	d	1
J ₁				
1	d	d	d	d

	xy ₂ 00	01	11	10
y ₁				
0	d	d	d	d
K ₁				
1	1	1	0	0

$$J_2 = xy_1,$$

$$J_1 = x,$$

$$Z = xy_2$$

$$k_2 = \bar{x}$$

$$k_1 = \bar{x}$$

- Draw the logic circuit diagram

$$\begin{aligned} J_2 &= xy_1, & k_2 &= \bar{x} \\ J_1 &= x, & k_1 &= \bar{x} \\ Z &= xy_2 \end{aligned}$$

